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Medical apparatus

Abstract

A medical apparatus includes a table portion that has a contact surface with which a subject comes into contact, a structural portion in which a relative position with the table portion is determined within a predetermined range, and a sterilization unit that is attached to the table portion or the structural portion and sterilizes the contact surface. The sterilization unit is configured to switch between a first state in which the sterilization unit is in contact with or close to the contact surface and a second state in which the sterilization unit is separated from the contact surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/JP2021/036240, filed Sep. 30, 2021, the disclosure of which is incorporated herein by reference in its entirety. Further, this application claims priority from Japanese Patent Application No. 2020-180888, filed on Oct. 28, 2020, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(1) The disclosed technology relates to a medical apparatus.

2. Description of the Related Art

(2) There is a medical apparatus such as a diagnosis apparatus and a treatment apparatus that sterilizes a portion of an examination table or the like that comes into contact with a subject is known. For example, JP2012-511381A discloses that, in an X-ray apparatus, an illumination device mounted on a C-arm sterilizes an examination table by transmitting UV radiation.

SUMMARY

(3) In a case where an examinee to be diagnosed or treated by a medical apparatus such as a diagnosis apparatus and a treatment apparatus is infected with a pathogen such as a bacterium or a virus, there is a risk that another examinee or a doctor will be infected with the pathogen via a portion of the medical apparatus with which the examinee comes into contact. Therefore, for example, it is preferable to increase the sterilization frequency such as sterilizing a portion of the medical apparatus with which the examinee comes into contact each time the treatment for one examinee has been completed.

(4) JP2012-511381A discloses that the illumination device mounted on the C-arm sterilizes the examination table by emitting ultraviolet rays. However, in a case where a distance from a light source that emits ultraviolet rays to the examination table is long, the energy of the ultraviolet rays is diffused, and thus a sufficient bactericidal effect cannot be obtained. On the other hand, in a case where the distance from the light source that emits ultraviolet rays to the examination table is short, there is a possibility that the normal use (for example, X-ray imaging) of the medical apparatus may be hindered.

(5) The disclosed technology has been made in view of the circumstances, and an object thereof is to perform, in a medical apparatus having a contact surface with which a subject comes into contact, effective sterilization of the contact surface without hindering normal use of the medical apparatus.

(6) According to the disclosed technology, there is provided a medical apparatus including a table portion that has a contact surface with which a subject comes into contact; a structural portion in which a relative position with the table portion is determined within a predetermined range; and a sterilization unit that is attached to the table portion or the structural portion and sterilizes the contact surface, in which the sterilization unit is configured to switch between a first state in which the sterilization unit is in contact with or close to the contact surface and a second state in which the sterilization unit is separated from the contact surface.

(7) It is preferable that in the first state, a distance between the sterilization unit and the contact surface may be 10 cm or less. The structural portion may be an examination unit that examines the subject in a non-contact manner. The structural portion may be a radiation source unit that emits radiation. The structural portion may be a detection unit that detects radiation. The sterilization unit may include a light source that emits ultraviolet rays.

- (8) The medical apparatus may further include a radiation source unit that emits radiation. In this case, the sterilization unit may include a phosphor that absorbs radiation and emits ultraviolet rays, and, in the first state, the radiation radiated from the radiation source unit may be applied to the phosphor.
- (9) The sterilization unit may eject a liquid or a gas having a bactericidal action toward the contact surface. The sterilization unit may include a wiping member that is a member impregnated with a liquid having a bactericidal action and wipes the contact surface.
- (10) The medical apparatus may further include a fixing unit that fixes the subject by sandwiching the subject between the table portion and the fixing unit. In this case, the sterilization unit may be configured to emit at least one of light, a liquid, or a gas having a bactericidal action from both a first surface and a second surface opposite to the first surface, and, in the first state, the first surface may face the contact surface of the table portion and the second surface may face a surface of the fixing unit that comes into contact with the subject.
- (11) The medical apparatus may further include a control unit that controls a sterilization process of sterilizing the contact surface with the sterilization unit. At least one of the sterilization unit or the contact surface may be movable in a surface direction of the contact surface. In this case, in the sterilization process, the control unit may set a state of the sterilization unit to the first state, and move at least one of the sterilization unit or the contact surface in the surface direction.
- (12) The medical apparatus may further include a region detection unit that detects a contact region of the contact surface with which the subject comes into contact. In this case, in the sterilization process, the control unit may sterilize the contact surface by setting a sterilization intensity in the contact region detected by the region detection unit to be higher than sterilization intensity in a region other than the contact region.
- (13) The medical apparatus may further include a height detection unit that detects a height of the contact surface. In this case, in the sterilization process, the control unit may control a height of the sterilization unit in accordance with a change in the height of the contact surface detected by the height detection unit such that a distance between the sterilization unit and the contact surface is constant.
- (14) The control unit may start the sterilization process on the basis of order information regarding diagnosis or treatment of the subject.
- (15) In the sterilization process, the control unit may sterilize the contact surface in a sterilization mode selected from a plurality of sterilization modes in which sterilization intensities are different from each other.
- (16) The table portion may be an examination table on which an examinee to be treated in the medical apparatus lies.
- (17) According to the disclosed technology, in a medical apparatus having a contact surface with which a subject comes into contact, it is possible to effectively sterilize the contact surface without hindering normal use of the medical apparatus.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Exemplary embodiments according to the technique of the present disclosure will be described in detail based on the following figures, wherein:
- (2) FIG. 1A is a diagram showing an example of a configuration of a CT apparatus that is an example of a medical apparatus according to an embodiment of the disclosed technology;
- (3) FIG. 1B is a view of the CT apparatus according to the embodiment of the disclosed technology as viewed from a rear surface side;
- (4) FIG. 2 is a diagram showing an example of a configuration of a sterilization unit according to

the embodiment of the disclosed technology;

(5) FIG. 3 is a flowchart showing an example of a flow of a sterilization process performed by a control unit according to the embodiment of the disclosed technology;

(6) FIG. 4 is a diagram showing an operation of a bed portion during the sterilization process according to the embodiment of the disclosed technology;

(7) FIG. 5A is a diagram showing another example of the configuration of the sterilization unit according to the embodiment of the disclosed technology;

(8) FIG. 5B is a diagram showing still another example of the configuration of the sterilization unit according to the embodiment of the disclosed technology;

(9) FIG. 5C is a diagram showing still another example of the configuration of the sterilization unit according to the embodiment of the disclosed technology;

(10) FIG. 6 is a view of a CT apparatus according to the embodiment of the disclosed technology as viewed from the rear surface side;

(11) FIG. 7A is a diagram showing the sterilization unit including a height detection unit according to the embodiment of the disclosed technology;

(12) FIG. 7B is a diagram showing the sterilization unit including the height detection unit according to the embodiment of the disclosed technology;

(13) FIG. 8 is a diagram showing a CT apparatus including a region detection unit according to the embodiment of the disclosed technology;

(14) FIG. 9A is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(15) FIG. 9B is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(16) FIG. 9C is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(17) FIG. 10A is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(18) FIG. 10B is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(19) FIG. 10C is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(20) FIG. 10D is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(21) FIG. 11 is a diagram showing a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(22) FIG. 12A is a diagram showing an example of a positioning mechanism of the sterilization unit including a phosphor according to the embodiment of the disclosed technology;

(23) FIG. 12B is a diagram showing an example of the positioning mechanism of the sterilization unit including the phosphor according to the embodiment of the disclosed technology;

(24) FIG. 12C is a diagram showing an example of the positioning mechanism of the sterilization unit including the phosphor according to the embodiment of the disclosed technology;

(25) FIG. 12D is a diagram showing an example of the positioning mechanism of the sterilization unit including the phosphor according to the embodiment of the disclosed technology;

(26) FIG. 12E is a diagram showing an example of the positioning mechanism of the sterilization unit including the phosphor according to the embodiment of the disclosed technology;

(27) FIG. 12F is a diagram showing an example of the positioning mechanism of the sterilization unit including the phosphor according to the embodiment of the disclosed technology;

(28) FIG. 13 is a flowchart showing an example of a flow of a sterilization process performed by the control unit according to the embodiment of the disclosed technology;

(29) FIG. 14 is a diagram showing an example of a configuration of a CT apparatus including a

second sterilization unit according to the embodiment of the disclosed technology;

(30) FIG. 15A is a sectional view taken along the line 15-15 in FIG. 14;

(31) FIG. 15B is a sectional view taken along the line 15-15 in FIG. 14;

(32) FIG. 15C is a sectional view taken along the line 15-15 in FIG. 14;

(33) FIG. 16A is a diagram showing an example of a configuration of the second sterilization unit according to the embodiment of the disclosed technology;

(34) FIG. 16B is a diagram showing an example of a configuration of the second sterilization unit according to the embodiment of the disclosed technology;

(35) FIG. 16C is a diagram showing an internal structure of the second sterilization unit according to the embodiment of the disclosed technology;

(36) FIG. 17A is a diagram showing an example of a configuration of a radiography apparatus 10A that is an example of a medical apparatus according to an embodiment of the disclosed technology;

(37) FIG. 17B is a view of the radiography apparatus according to the embodiment of the disclosed technology as viewed from a direction of an arrow A in FIG. 17A;

(38) FIG. 18A is a plan view corresponding to FIG. 17A, showing a state of the sterilization unit in a case where a contact surface is sterilized;

(39) FIG. 18B is a plan view corresponding to FIG. 17B, showing a state of the sterilization unit in a case where the contact surface is sterilized;

(40) FIG. 19 is a diagram showing an example of a configuration of a radiography apparatus that is an example of a medical apparatus according to an embodiment of the disclosed technology;

(41) FIG. 20A is a diagram showing a state of the sterilization unit in a case where a contact surface is sterilized in the radiography apparatus according to the embodiment of the disclosed technology;

(42) FIG. 20B is a diagram showing a state of the sterilization unit in a case where the contact surface is sterilized in the radiography apparatus according to the embodiment of the disclosed technology;

(43) FIG. 21A is a diagram showing an example of a configuration of a mammography apparatus that is an example of a medical apparatus according to an embodiment of the disclosed technology;

(44) FIG. 21B is a view of the mammography apparatus according to the embodiment of the disclosed technology as viewed from a direction of an arrow B in FIG. 21A;

(45) FIG. 22A is a plan view corresponding to FIG. 21A, showing a state of the sterilization unit in a case where the contact surface is sterilized;

(46) FIG. 22B is a plan view corresponding to FIG. 21B showing a state of the sterilization unit in a case where the contact surface is sterilized;

(47) FIG. 23 is a diagram showing an example of a configuration of the sterilization unit according to the embodiment of the disclosed technology;

(48) FIG. 24A is a diagram showing an example of a configuration of the sterilization unit according to the embodiment of the disclosed technology; and

(49) FIG. 24B is a sectional view taken along the line 24B-24B in FIG. 24A.

DETAILED DESCRIPTION

(50) Hereinafter, an example of an embodiment of the disclosed technology will be described with reference to the drawings. In each drawing, the same or equivalent constituents and portions are given the same reference numerals, and redundant description will be omitted as appropriate.

First Embodiment

(51) FIG. 1A is a diagram showing an example of a configuration of a CT apparatus 10 that is an example of a medical apparatus according to a first embodiment of the disclosed technology. The CT apparatus 10 according to the present embodiment includes a gantry 11 and an examination table 16. The gantry 11 has a tunnel-like structure having an opening portion 12 at the center thereof. A radiation source unit 13 that emits X-rays and a detection unit 14 that detects X-rays and generates a radiation image are provided inside the gantry 11. The radiation source unit 13 and the detection unit 14 can be each rotated along the annular shape of the gantry 11 while maintaining a

positional relationship of facing each other. A control unit **15** that controls an operation of the CT apparatus **10** is provided inside the gantry **11**.

(52) The examination table **16** has a base portion **16A** fixed to a floor and a bed portion **16B** on which a subject (examinee) lies. A surface of the bed portion **16B** is a contact surface **17** with which the subject (examinee) comes into contact. The bed portion **16B** is slidable in a surface direction of the contact surface **17**. In a case of capturing a radiation image, the bed portion **16B** is slid to transport a subject (examinee) lying on the bed portion **16B** into the opening portion **12** of the gantry **11**.

(53) The examination table **16** is an example of a “table portion” in the disclosed technology. The “table portion” has a contact surface with which a subject (examinee) comes into contact. The gantry **11** is an example of a “structural portion” in the disclosed technology, and is also an example of an “examination unit”. The “structural portion” is defined as a relative position with respect to the “table portion” within a predetermined range. The “examination unit” examines a subject (examinee) in a non-contact manner. In the following description, a vertical direction will be referred to as a Z direction, a slide direction of the bed portion **16B** will be referred to as a Y direction, and a direction perpendicular to both the Z direction and the Y direction will be referred to as an X direction.

(54) FIG. 1B is a view (X-Z plan view) of the CT apparatus **10** viewed from a rear surface side (a side opposite to the installation side of the examination table **16**). A sterilization unit **20** that sterilizes the contact surface **17** is attached to a rear surface of the gantry **11**. The sterilization unit **20** has an elongated rod-like shape, and one end thereof is attached to the gantry **11**. A rotation shaft **18** with the Y direction as an axial direction is provided in the attachment portion of the sterilization unit **20**, and the sterilization unit **20** is rotatable about an axis of the rotation shaft **18**.

(55) In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into a first state of being brought into contact with or close to the contact surface **17** (a state in which a sterilization process can be performed). That is, as indicated by a solid line in FIGS. 1A and 1B, the sterilization unit **20** is positioned such that a longitudinal direction thereof is along the X direction to cross the opening portion **12** of the gantry **11**. On the other hand, in a case where a radiation image is captured by the CT apparatus **10**, that is, in a normal use of the CT apparatus **10**, the sterilization unit **20** is brought into a second state (retracted state) of being separated from the contact surface **17**. That is, as shown by the dotted lines in FIGS. 1A and 1B, the sterilization unit **20** is positioned such that the longitudinal direction thereof is along the Y direction so as not to hinder the movement of the bed portion **16B**.

(56) FIG. 2 is a diagram showing an example of a configuration of the sterilization unit **20**. The sterilization unit **20** includes a plurality of light sources **22** that are arranged on a wiring board **21** and emit light having a bactericidal action. The contact surface **17** of the examination table **16** is irradiated with light from the light source **22** and thus the contact surface **17** can be sterilized. As the light source **22**, for example, a light emitting diode (LED) that emits ultraviolet rays (UV-C) having a wavelength of 200 to 280 nm may be used. For example, by applying ultraviolet rays having a wavelength of 265 nm at 1.7 mJ/cm^{sup.2}, it is possible to kill 99.9% of the human coronavirus. A mercury lamp may also be used as the light source **22** that emits ultraviolet rays.

(57) A length L of the sterilization unit **20** in the longitudinal direction is preferably the equal to or less than a length of the contact surface **17** in the X direction. The sterilization unit **20** may include a lens that collects light from the light source **22**. However, in a case where a UV-C wave is used as light having a bactericidal action, a material of the lens is limited to a material (for example, quartz and sapphire) that can transmit the UV-C wave therethrough, and thus the cost is high. In a case where a lens is not used, it is preferable that a distance between the sterilization unit **20** and the contact surface **17** is as short as possible. In order to perform effective sterilization, a distance D between the sterilization unit **20** and the contact surface **17** is preferably 10 cm or less, more preferably 5 cm or less, and most preferably 2 cm or less. In a case where the bed portion **16B** can

be lifted and lowered in the Z direction, the distance D between the sterilization unit **20** and the contact surface **17** may be adjusted by adjusting a height position of the bed portion **16B**.

(58) FIG. **3** is a flowchart showing an example of a flow of a sterilization process performed by the control unit **15** in a case where the contact surface **17** is sterilized. The sterilization process is executed, for example, in a case where a user performs an operation for starting the sterilization process. The operation for starting the sterilization process is performed by using, for example, an operation unit or a console (neither of that is shown) provided in the CT apparatus **10**.

(59) In step S**1**, the control unit **15** causes a state of the sterilization unit **20** to transition to the first state (a state in which the sterilization process can be performed). Consequently, the sterilization unit **20** is positioned such that the longitudinal direction thereof is along the X direction. In step S**2**, the control unit **15** turns on the light source **22** of the sterilization unit **20**. In step S**3**, the control unit **15** slides the bed portion **16B** of the examination table **16**.

(60) FIG. **4** is a diagram (Y-Z sectional view) showing an operation of the bed portion **16B** during the sterilization process. As shown in FIG. **4**, the control unit **15** slides the bed portion **16B** while maintaining the first state of the sterilization unit **20**. Consequently, regions of the contact surface **17** that pass directly under the sterilization unit **20** are moved sequentially, and thus it is possible to irradiate the entire contact surface **17** with light having a bactericidal action emitted from the sterilization unit **20**. In a case where the gantry **11** is movable along the Y direction, the gantry **11** may be moved together with the sterilization unit **20** in order to irradiate the entire contact surface **17** with the light from the sterilization unit **20**.

(61) In step S**4**, the control unit **15** determines whether or not sterilization of the contact surface **17** has been completed. In the control unit **15**, for example, in a case where the bed portion **16B** reaches a terminal position on the rear surface side of the gantry **11** and then returns to an initial position (that is, in a case where the bed portion **16B** reciprocates once), it is determined that the sterilization has been completed. In a case where the control unit **15** determines that the sterilization of the contact surface **17** has been completed, the process proceeds to step S**5**.

(62) In step S**5**, the control unit **15** turns off the light source **22**. In step S**6**, the control unit **15** causes a state of the sterilization unit **20** to transition to the second state (retracted state). Consequently, the control unit **15** positions the sterilization unit **20** such that the longitudinal direction thereof is along the Y direction.

(63) As described above, in the CT apparatus **10** according to the present embodiment, the first state in which the sterilization unit **20** is in contact with or close to the contact surface **17** and the second state in which the sterilization unit **20** is separated from the contact surface **17** can switch therebetween. In a case where the contact surface **17** is sterilized, by setting a state of the sterilization unit **20** to the first state, it is possible to effectively sterilize the contact surface **17**. In a case where the contact surface **17** is not sterilized, by setting a state of the sterilization unit **20** to the second state, it is possible to avoid hindering the normal use of the CT apparatus **10** (that is, capturing a radiation image). That is, according to the CT apparatus **10** according to the present embodiment, it is possible to effectively sterilize the contact surface **17** without hindering the normal use of the CT apparatus **10**.

(64) In the above description, as an example, the sterilization unit **20** includes the light source **22** that emits light having a bactericidal action such as ultraviolet rays, but the present invention is not limited to this aspect. FIGS. **5A**, **5B**, and **5C** are diagrams (Y-Z plan view) respectively showing other examples of a configuration of the sterilization unit **20**.

(65) As shown in FIG. **5A**, the sterilization unit **20** may eject a gas or a liquid having a bactericidal action toward the contact surface **17**. As the gas having a bactericidal action, for example, ozone or plasma may be used. As the liquid having a bactericidal action, for example, alcohol or a hypochlorous acid solution may be used. The sterilization unit **20** may be one that sprays aerosol that is a solid or liquid particle having a bactericidal action, dispersed in a gas.

(66) As shown in FIG. **5B**, the sterilization unit **20** may include a wiping member **23** that is a

member impregnated with a liquid having a bactericidal action such as alcohol or a hypochlorous acid solution and wipes the contact surface **17**. As the wiping member **23**, a soft member such as a cloth or a sponge may be used.

(67) As shown in FIG. 5C, the sterilization unit **20** may have a configuration in which a plurality of types of sterilization members different from each other are arranged along the Y direction that is the slide direction of the bed portion **16B**. In the example shown in FIG. 5C, a configuration in which a light source **22** that emits light having a bactericidal action and a wiping member **23** impregnated with a liquid having a bactericidal action are arranged along the Y direction is exemplified. By arranging a plurality of types of sterilization members different from each other in the slide direction of the bed portion **16B**, it is possible to promote the bactericidal action in the sterilization unit **20**. A combination of a plurality of types of sterilization members may be selected as appropriate, and three or more types of sterilization members may be combined.

(68) In the above description, a case where switching between the first state and the second state is performed by rotating the sterilization unit **20** about the axis of the rotation shaft **18** has been described as an example, but the present invention is limited to this aspect. For example, as shown in FIG. 6, switching between the first state and the second state may be performed by lifting and lowering the sterilization unit **20** in the Z direction along guide rods **31**. In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into the first state of being in contact with or close to the contact surface **17** (a state in which the sterilization process can be performed), and is positioned at a lowered position as indicated by a solid line in FIG. 6. On the other hand, in a case where a radiation image is captured by the CT apparatus **10**, that is, in the normal use of the CT apparatus **10**, the sterilization unit **20** is brought into the second state (retracted state) of being separated from the contact surface **17**, and is positioned at a lifted position as indicated by a dashed line in FIG. 6.

(69) As shown in FIG. 7A, the CT apparatus **10** may include a height detection unit **24** that detects a height of the contact surface **17**. As the height detection unit **24**, for example, a TOF camera that captures a distance image indicating a distance to an imaging target by using a time of flight (TOF) method may be used.

(70) In a case where the CT apparatus **10** includes the height detection unit **24**, as shown in FIG. 7B, the control unit **15** may lift and lower the sterilization unit **20** in accordance with a change in a height of the contact surface **17** detected by the height detection unit **24** such that a distance between the sterilization unit **20** and the contact surface **17** is constant in the sterilization process. Consequently, even in a case where a structure such as a pillow is installed on the examination table **16** and the contact surface **17** has irregularities, a distance between the sterilization unit **20** and the contact surface **17** is kept constant, and thus the contact surface **17** can be appropriately sterilized. Although FIG. 7A and FIG. 7B exemplify a configuration in which the height detection unit **24** is attached to the sterilization unit **20**, the height detection unit **24** may be attached to the opening portion **12** of the gantry **11**.

(71) In the above description, a case where the sterilization process is started on the basis of an operation of a user has been described as an example, but the sterilization process may be started on the basis of order information issued before a radiation image is captured by the CT apparatus **10**. For example, in a case where the order information includes information related to a radiation image capturing schedule, the control unit **15** may start the sterilization process on the basis of the information related to the capturing schedule included in the order information. The control unit **15** may specify, for example, a timing after the end of imaging of a certain examinee and before the start of imaging of the next examinee, and start the sterilization process at that timing.

(72) In the sterilization process, the control unit **15** may sterilize the contact surface **17** in a sterilization mode selected from a plurality of sterilization modes in which sterilization intensities are different from each other. For example, in a case where the sterilization unit **20** includes the light source **22** that emits ultraviolet rays, a sterilization intensity may be changed by changing at

least one of an irradiation time, an irradiation range, or an irradiation intensity of the ultraviolet rays.

(73) The irradiation time of the ultraviolet rays may be changed depending on, for example, a movement speed of the bed portion **16B** that is being slid during the sterilization process. The irradiation time of the ultraviolet rays may also be changed depending on the number of times of reciprocation of the bed portion **16B** during the sterilization process. The irradiation range of the ultraviolet rays may be changed depending on a movement range of the bed portion **16B** that is being slid during the sterilization process. The irradiation intensity of the ultraviolet rays may be changed depending on, for example, a driving current of the light source **22** that emits the ultraviolet rays. The sterilization mode is selected through, for example, a user operation. A sterilization mode selection operation is performed by using, for example, an operation unit or a console (neither of that is shown) provided in the CT apparatus **10**. As the sterilization mode, for example, three modes such as a standard mode, a high-speed mode, and an enhancement mode may be provided.

(74) In a case where the standard mode is selected, in the sterilization process, the control unit **15** sets a movement speed of the bed portion **16B** to a standard speed, and causes the bed portion **16B** to reciprocate once to irradiate the entire contact surface **17** with ultraviolet rays.

(75) In a case where the high-speed mode is selected, the control unit **15** sets a movement speed of the bed portion **16B** to a speed higher than the standard speed in the sterilization process and restricts a movement range of the contact surface **17** to irradiate only a partial region of the contact surface **17** with ultraviolet rays. Consequently, a sterilization intensity is lower than in the standard mode, but the processing time is reduced. The high-speed mode is, for example, a mode that can be selected in a case where a large number of imaging orders are in a standby state.

(76) In a case where the enhancement mode is selected, the control unit **15** sets a movement speed of the bed portion **16B** to a speed slower than the standard speed in the sterilization process, and causes the bed portion **16B** to reciprocate a plurality of times such that a process of irradiating the entire contact surface **17** with ultraviolet rays is repeated a plurality of times. Consequently, although the processing time is longer than that in the standard mode, a sterilization intensity is increased, and more reliable sterilization can be performed. The enhancement mode is, for example, a mode that can be selected in a case where the CT apparatus **10** is used for an examinee suspected of being infected with a pathogen.

(77) As shown in FIG. **8**, the CT apparatus **10** may include a region detection unit **19** that detects a contact region of contact surface **17** with which a subject (examinee) comes into contact. As the region detection unit **19**, for example, a visible light camera installed such that the entire contact surface **17** is within an imaging visual field may be used. In a case where the CT apparatus **10** includes the region detection unit **19**, the control unit **15** may set a sterilization intensity in a contact region detected by the region detection unit **19** in the sterilization process to be higher than that in a region other than the contact region, and sterilize the contact surface **17**. For example, the control unit **15** may set a movement speed of the bed portion **16B** in a case where the contact region passes directly under the sterilization unit **20** to be lower than a movement speed of the bed portion **16B** in a case where a region other than the contact region passes directly under the sterilization unit **20**. The control unit **15** may set an irradiation intensity of ultraviolet rays in a case where the contact region passes directly under the sterilization unit **20** to be higher than an irradiation intensity of ultraviolet rays in a case where a region other than the contact region passes directly under the sterilization unit **20**. As described above, the contact surface **17** is sterilized by setting the sterilization intensity in the contact region to be higher than that in the region other than the contact region, and thus the sterilization process can be efficiently performed.

(78) In the above description, the configuration in which the sterilization unit **20** is attached to the gantry **11** has been described as an example. However, as shown in FIG. **9A**, the sterilization unit **20** may be attached to the examination table **16**. In the example shown in FIG. **9A**, the sterilization

unit **20** is attached to the base portion **16A** of the examination table **16** via an arm portion **32**. A rotation shaft **33** having the Y direction as an axial direction is provided between the sterilization unit **20** and the arm portion **32** in the state shown in FIG. **9A**, and the sterilization unit **20** is rotatable about an axis of the rotation shaft **33**. The arm portion **32** is rotatable about an axis of the rotation shaft **34** having the X direction as an axial direction.

(79) In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into a first state of being brought into contact with or close to the contact surface **17** (a state in which a sterilization process can be performed). That is, as shown in FIG. **9A**, the arm portion **32** is positioned such that the longitudinal direction thereof is along the Z direction, and the sterilization unit **20** is positioned such that the longitudinal direction thereof is along the X direction. As shown in FIG. **9B**, by sliding the bed portion **16B** while setting a state of the sterilization unit **20** to the first state, it is possible to sterilize the entire contact surface **17**. On the other hand, in a case where a radiation image is captured by the CT apparatus **10**, that is, in a normal use of the CT apparatus **10**, the sterilization unit **20** is brought into a second state (retracted state) of being separated from the contact surface **17**. That is, as shown in FIG. **9C**, the sterilization unit **20** is positioned such that the longitudinal direction thereof is along the longitudinal direction of the arm portion **32**, and the arm portion **32** is positioned such that the longitudinal direction thereof is along the Y direction.

(80) As described above, even in a configuration in which the sterilization unit **20** is attached to the examination table **16**, the contact surface **17** can be effectively sterilized without hindering the normal use of the CT apparatus **10**.

(81) FIG. **10A** is a diagram showing another configuration example of the sterilization unit **20** attached to the examination table **16**. FIG. **10B** is a sectional view taken along the line **10B-10B** in FIG. **10A**. The sterilization unit **20** is attached to the examination table **16** via a pair of arm portions **35** attached to both ends in the longitudinal direction thereof. A roller **36** is provided at one end of the arm portion **35** in the longitudinal direction. The roller **36** is fitted in a guide groove **37** provided in the Y direction on the side surface of the bed portion **16B**. By driving the roller **36** with a motor (not shown), the arm portions **35** can be moved together with the sterilization unit **20** in the Y direction. The arm portion **35** is rotatable about an axis of a rotation shaft **38** provided at the same position as the attachment position of the roller **36** with the X direction as an axial direction.

(82) In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into a first state of being brought into contact with or close to the contact surface **17** (a state in which a sterilization process can be performed). That is, as shown in FIGS. **10A** and **10B**, the arm portion **35** is positioned such that the sterilization unit **20** is located above the bed portion **16B**. As shown in FIG. **10C**, by driving the roller **36** while setting a state of the sterilization unit **20** to the first state, the arm portion **35** is moved together with the sterilization unit **20** in the extension direction (Y direction) of the guide groove **37**. Consequently, it is possible to sterilize the entire contact surface **17**. On the other hand, in a case where a radiation image is captured by the CT apparatus **10**, that is, in a normal use of the CT apparatus **10**, the sterilization unit **20** is brought into a second state (retracted state) of being separated from the contact surface **17**. That is, as shown in FIG. **10D**, the arm portion **35** is positioned such that the sterilization unit **20** is located below the bed portion **16B**.

(83) As described above, even in a configuration in which the sterilization unit **20** is self-propelled, the contact surface **17** effectively sterilized without hindering the normal use of the CT apparatus **10**. Even in a case where the CT apparatus **10** does not include a mechanism for sliding the bed portion **16B**, it is possible to sterilize the entire contact surface **17** by applying this configuration.

(84) FIG. **11** is a diagram showing another configuration of the sterilization unit **20**. The sterilization unit **20** may include a phosphor **25** that absorbs radiation (X-rays) and emits ultraviolet rays. As radiation applied to the phosphor **25**, X-rays radiated from the radiation source unit **13** are used. As the phosphor **25**, for example, Cs.sub.2ZnCl.sub.4, HfP.sub.2O.sub.7, BaFCl:Eu, BaFBr:Eu, YTaO.sub.4, or Ba.sub.2O.sub.2S:Tb may be used.

(85) FIG. 12A is a diagram (X-Z plan view) showing an example of a positioning mechanism of the sterilization unit **20** including the phosphor **25**. FIG. 12B is a sectional view (Y-Z sectional view) along the line 12B-12B in FIG. 12A, and FIG. 12C is a sectional view (X-Y sectional view) along the line 12C-12C in FIG. 12A.

(86) As shown in FIGS. 12A to 12C, the sterilization unit **20** including the phosphor **25** is attached to the gantry **11** via a support portion **40**, a first guide rail **41**, and a second guide rail **42**. The second guide rail **42** includes a pair of rod-shaped members having the Z direction as a longitudinal direction, and is fixed to the rear surface of the gantry **11**. The first guide rail **41** includes a pair of rod-shaped members of which a longitudinal direction is the Y direction, and is slidably provided at the second guide rail **42**. That is, the first guide rail **41** is movable (liftable and lowerable) in the Z direction. The support portion **40** is a member that supports the sterilization unit **20** at an end portion on the back side of the sterilization unit **20** (left side in FIG. 12B) and is slidably provided at the first guide rail **41**. That is, the support portion **40** can be moved in the Y direction together with the sterilization unit **20**.

(87) In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into a first state of being brought into contact with or close to the contact surface **17** (a state in which a sterilization process can be performed). That is, as shown in FIG. 12B, the first guide rail **41** is positioned at a height position near the center of the opening portion **12** of the gantry **11**, and the support portion **40** is positioned at the terminal position on the front side of the first guide rail **41** (right side in FIG. 12B). Consequently, the sterilization unit **20** is disposed in the opening portion **12** of the gantry **11**. On the other hand, the radiation source unit **13** is positioned such that the phosphor **25** is included in an irradiation field of X-rays. That is, the radiation source unit **13** is disposed directly above the phosphor **25**.

(88) In this state, by radiating X-rays from the radiation source unit **13**, the X-rays are applied to the phosphor **25**, and ultraviolet rays are emitted from the phosphor **25**. The ultraviolet rays emitted from the phosphor **25** are applied to the contact surface **17** of the bed portion **16B**. As shown in FIG. 12B, by sliding the bed portion **16B** while setting a state of the sterilization unit **20** to the first state, it is possible to sterilize the entire contact surface **17**. During the sterilization process, X-rays are continuously radiated from the radiation source unit **13**.

(89) On the other hand, in a case where a radiation image is captured by the CT apparatus **10**, that is, in a normal use of the CT apparatus **10**, the sterilization unit **20** is brought into a second state (retracted state) of being separated from the contact surface **17**. That is, in a case where the sterilization process has been completed, as shown in FIG. 12D, the support portion **40** is moved to the terminal position on the back side of the first guide rail **41** (left side in FIG. 12D). Thereafter, as shown in FIG. 12E, the first guide rail **41** is moved to the upper end position of the second guide rail **42**, and then, as shown in FIG. 12F, the support portion **40** is moved to the terminal position on the front side of the first guide rail **42** (right side in FIG. 12F). Consequently, the sterilization unit **20** is disposed above the gantry **11**.

(90) FIG. 13 is a flowchart showing an example of a flow of the sterilization process performed by the control unit **15** in a case where the contact surface **17** is sterilized in the CT apparatus **10** having the positioning mechanism of the sterilization unit **20** including the phosphor **25** described above.

(91) In step S11, the control unit **15** causes a state of the sterilization unit **20** to transition to the first state shown in FIG. 12B. Consequently, the sterilization unit **20** including the phosphor **25** is disposed in the opening portion **12** of the gantry **11**.

(92) In step S12, the control unit **15** positions the radiation source unit **13** such that the phosphor **25** is included in the irradiation field of X-rays radiated from the radiation source unit **13**. That is, the radiation source unit **13** is disposed directly above the phosphor **25**.

(93) In step S13, the control unit **15** causes the radiation source unit **13** to start to radiate X-rays. Consequently, the X-rays from the radiation source unit **30** are applied to the phosphor **25**, and the phosphor **25** emits ultraviolet rays. The ultraviolet rays emitted from the phosphor **25** are applied to

the contact surface **17**. In step **S14**, the control unit **15** slides the bed portion **16B** of the examination table **16**. Consequently, the entire contact surface **17** is irradiated with the ultraviolet rays. During the period in which the bed portion **16B** is being slid, X-rays are continuously emitted from the radiation source unit **13**.

(94) In step **S15**, the control unit **15** determines whether or not the sterilization of the contact surface **17** has been completed. In the control unit **15**, for example, in a case where the bed portion **16B** reaches a terminal position on the rear surface side of the gantry **11** and then returns to an initial position (that is, in a case where the bed portion **16B** reciprocates once), it is determined that the sterilization has been completed. In a case where the control unit **15** determines that the sterilization of the contact surface **17** has been completed, the process proceeds to step **S16**.

(95) In step **S16**, the control unit **15** stops radiating the X-rays from the radiation source unit **13**. In step **S16**, a state of the sterilization unit **20** transitions to the second state (retracted state).

Consequently, as shown in FIG. **12F**, the sterilization unit **20** is disposed above the gantry **11**.

(96) As described above, even in a configuration in which the sterilization unit **20** includes the phosphor **25**, the contact surface **17** can be effectively sterilized without hindering the normal use of the CT apparatus **10**. As a configuration other than the configuration in which the sterilization unit including the phosphor **25** is configured with the guide rails **41** and **42**, the sterilization unit may be configured with the arm as shown in FIGS. **10A** to **10D**. For sterilization, in a case where the bed portion **16B** performs sterilization while traveling in the direction of the gantry ($-Y$ direction) at a speed V , the arm having the sterilization unit **20** is moved in advance to the end of the bed portion **16B** near the gantry **11**. Next, the bed portion **16B** is moved in the direction of the gantry **11** at the speed V for sterilization, and from a time at which the arm reaches the optimum position where X-rays are applied, which is the center position of the gantry, the arm is stopped at that position by moving the arm in the direction opposite to the gantry **11** ($+Y$ direction) at the same speed V as that of the bed portion **16B**, that is, by controlling a movement speed of the bed portion **16B** and a movement speed of the arm to be the same and the traveling directions to be opposite to each other. Thus, the sterilization unit is constantly irradiated with X-rays, and the phosphor **25** that has absorbed the X-rays emits ultraviolet rays such that sterilization can be performed.

(97) FIG. **14** is a diagram (X-Z plan view) showing an example of a configuration of the CT apparatus **10** including a second sterilization unit **20B** that sterilizes an inner wall surface **12A** of the opening portion **12** of the gantry **11**. FIGS. **15A**, **15B**, and **15C** each are sectional views (X-Y sectional views) taken along the line **15-15** in FIG. **14**.

(98) The second sterilization unit **20B** is attached to a first sterilization unit **20A** that sterilizes the contact surface **17** of the examination table **16**. The first sterilization unit **20A** corresponds to the sterilization unit **20** shown in FIGS. **1A** and **1B**. The second sterilization unit **20B** is attached to the first sterilization unit **20A** via an arm portion **26**. The second sterilization unit **20B** is attached to a distal end of the arm portion **26**, and a basal end of the arm portion **26** is attached to the center of the first sterilization unit **20A** in the longitudinal direction. A rotation shaft **27** having the Y direction as an axial direction is provided in the attachment part of the arm portion **26** to the first sterilization unit **20A**, and the arm portion **26** is rotatable by 360° about an axis of the rotation shaft **27**.

(99) FIGS. **16A** and **16B** are diagrams showing an example of a configuration of the second sterilization unit **20B**, and FIG. **16C** is a diagram showing an internal structure of the second sterilization unit **20B**. The second sterilization unit **20B** includes a plurality of tubular members **28** having diameters different from each other, and a light source **22B** that is provided inside the tubular member **28** and emits light having a bactericidal action such as ultraviolet rays. The second sterilization unit **20B** is extendable and contractible, and in the extended state shown in FIG. **16A**, the plurality of tubular members **28** are disposed such that the diameters gradually decrease toward the distal end. On the other hand, in the contracted state shown in FIG. **16B**, the tubular member **28**

having a relatively small diameter is accommodated inside the tubular member **28** having a relatively large diameter.

(100) An opening portion **29** is provided in each of the tubular members **28**. An inner wall surface of the tubular member **28** is covered with a reflective film made of a member having light reflectivity such as aluminum. Consequently, as shown in FIG. **16C**, the light emitted from the light source **22B** is reflected by the inner wall surface of the tubular member **28** and emitted from the opening portion **29** to the outside of the tubular member **28**. The opening portion **29** is disposed in a direction facing the inner wall surface **12A** of the opening portion **12** of the gantry **11**.

(101) In a case where the inner wall surface **12A** is sterilized by the second sterilization unit **20B**, as shown in FIGS. **15A**, **15B**, and **16A**, the second sterilization unit **20B** is brought into an extended state, and the arm portion **26** is rotated about the axis of the rotation shaft **27**.

Consequently, as shown in FIGS. **15A** and **15B**, the second sterilization unit **20B** is moved along the inner wall surface **12A** while maintaining a state of being in contact with or close to the inner wall surface **12A**. Consequently, the light emitted from the light source **22B** is applied to the entire inner wall surface **12A**.

(102) On the other hand, in a case where a radiation image is captured by the CT apparatus **10**, that is, in a normal use of the CT apparatus **10**, as shown in FIGS. **15C** and **16B**, the second sterilization unit **20B** is brought into a contracted state. Thereafter, as indicated by the dotted line in FIG. **14**, the first sterilization unit **20A** is positioned such that the longitudinal direction thereof is along the Z direction.

(103) As described above, by providing the CT apparatus **10** with the second sterilization unit **20B** in addition to the first sterilization unit **20A**, not only the contact surface **17** of the examination table **16** but also the inner wall surface **12A** of the opening portion **12** of the gantry **11** can also be sterilized. By making the arm portion **26** provided with the second sterilization unit **20B** rotatable about the axis of the rotation shaft **27**, it is possible to sterilize the entire inner wall surface **12A**. By making the second sterilization unit **20B** extendable and contractible, it is possible to perform the sterilization process without hindering the normal use of the CT apparatus **10**.

(104) In the above description, an aspect in which the second sterilization unit **20B** includes the light source **22B** that emits light having a bactericidal action has been described as an example, but the present invention is not limited to this aspect. For example, the second sterilization unit **20B** may be one that ejects a gas having a bactericidal action such as ozone or plasma or a liquid having a bactericidal action such as alcohol or a hypochlorous acid solution. The second sterilization unit **20B** may be a wiping member that is member impregnated with a liquid having a bactericidal action such as alcohol or a hypochlorous acid solution and wipes the inner wall surface **12A**. The second sterilization unit **20B** may have a configuration in which a plurality of types of sterilization members different from each other are arranged. A configuration in which sterilization is performed by using ultraviolet rays emitted by a phosphor (FIG. **12**) that absorbs X-rays may be used. Since directions of fluorescence emission are isotropic (360-degree omnidirectional), in a case of using a configuration in which light is emitted from both upper and lower surfaces of the inner wall surface of the opening portion of the gantry **11**, the inner wall of the gantry **11** can be sterilized, and the upper part of the inner wall surface (to which a virus is more likely to adhere) can also be sterilized at the same time as sterilization of the examination table **16**.

Second Embodiment

(105) FIG. **17A** is a diagram (X-Z plan view) showing an example of a configuration of a radiography apparatus **10A** that is an example of a medical apparatus according to a second embodiment of the disclosed technology. FIG. **17B** is a view (Y-Z plan view) of the radiography apparatus **10A** viewed from a direction of an arrow A in FIG. **17A**. In the following description, a vertical direction will be referred to as a Z direction, a slide direction of the bed portion **16B** will be referred to as a Y direction, and a direction perpendicular to both the Z direction and the Y direction will be referred to as an X direction.

(106) The radiography apparatus **10A** has a so-called C-arm type form, and includes a radiation source unit **13** attached to one end of a C-arm **50** and a detection unit **14** attached to the other end of the C-arm **50**. The radiography apparatus **10A** further includes a control unit **15** and an examination table **16**. The C-arm **50** is rotatable in a **0** direction and a **1** direction shown in FIG. **17A**, and is further liftable and lowerable in the Z direction. The radiation source unit **13** and the detection unit **14** can be each rotated and lifted and lowered in accordance with rotation and movement of the C-arm **50** while maintaining a positional relationship of facing each other. In the radiography apparatus **10A** according to the present embodiment, the detection unit **14** is disposed above a bed portion **16B**, and the radiation source unit **13** is disposed below the bed portion **16B**.

(107) The examination table **16** has a base portion **16A** fixed to a floor and a bed portion **16B** on which a subject (examinee) lies. A surface of the bed portion **16B** is a contact surface **17** with which the subject (examinee) comes into contact. The bed portion **16B** is installed to protrude into a region between the radiation source unit **13** and the detection unit **14** attached to the C-arm **50**. In a case of capturing a radiation image, by sliding the bed portion **16B**, it is possible to cause any part of a subject (examinee) to be included in an imaging visual field.

(108) The examination table **16** is an example of a “table portion” in the disclosed technology. The “table portion” has a contact surface with which a subject comes into contact. Each constituent of the structure including the C-arm **50**, the radiation source unit **13**, and the detection unit **14** is an example of a “structural portion” in the disclosed technology, and is also an example of an “examination unit”. The “structural portion” is defined as a relative position with respect to the “table portion” within a predetermined range. The “examination unit” examines a subject in a non-contact manner.

(109) A sterilization unit **20** that sterilizes the contact surface **17** of the bed portion **16B** is attached to the detection unit **14** via an arm portion **51**. The sterilization unit **20** has an elongated rod-like shape, and one end of the sterilization unit **20** in the longitudinal direction is connected to one end of the arm portion **51**. A rotation shaft **52** having the X direction as an axial direction is provided at the other end of the arm portion **51**, and the arm portion **51** is rotatable about an axis of the rotation shaft **52**. Similar to the first embodiment, as the sterilization unit **20**, a sterilization unit that emit light having a bactericidal action, a sterilization unit that ejects a gas or a liquid having a bactericidal action, a sterilization unit including a wiping member, and a sterilization unit having a configuration in which a plurality of types of different sterilization members are arranged may be employed.

(110) FIGS. **18A** and **18B** are diagrams showing a state of the sterilization unit **20** in a case where the contact surface **17** is sterilized, and are plan views corresponding to FIGS. **17A** and **17B**, respectively. In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into a first state of being brought into contact with or close to the contact surface **17** (a state in which a sterilization process can be performed). That is, as shown in FIGS. **18A** and **18B**, the arm portion **51** is positioned such that the sterilization unit **20** is located between the detection unit **14** and the bed portion **16B**. As shown in FIG. **18B**, by sliding the bed portion **16B** while setting a state of the sterilization unit **20** to the first state, it is possible to sterilize the entire contact surface **17**. A distance between the sterilization unit **20** and the contact surface **17** may be adjusted by adjusting a height position of the C-arm **50**. In a case where the bed portion **16B** is liftable and lowerable in the Z direction, a distance between the sterilization unit **20** and the contact surface **17** may be adjusted by adjusting a height position of the bed portion **16B**.

(111) On the other hand, in a case where a radiation image is captured by the radiography apparatus **10A**, that is, in the normal use of the radiography apparatus **10A**, the sterilization unit **20** is brought into the second state (retracted state) of being separated from the contact surface **17**. Consequently, as shown in FIGS. **17A** and **17B**, the arm portion **51** is positioned such that the sterilization unit **20** is located above the detection unit **14**.

(112) According to the radiography apparatus **10A** of the second embodiment of the disclosed

technology, in the same manner as in the first embodiment, the contact surface **17** can be effectively sterilized without hindering the normal use of the radiography apparatus **10A**.

(113) In the above description, the case where the detection unit **14** is disposed above the bed portion **16B** and the radiation source unit **13** is disposed below the bed portion **16B** has been described as an example. However, the radiation source unit **13** may be disposed above the bed portion **16B**, and the detection unit **14** may be disposed below the bed portion **16B**. In this case, the sterilization unit **20** is attached to the radiation source unit **13** disposed above the bed portion **16B** via the arm portion. In the radiography apparatus **10A**, the sterilization units **20** in the forms shown in FIGS. **9A** to **9C**, FIGS. **10A** to **10D**, and FIG. **11** may be employed.

Third Embodiment

(114) FIG. **19** is a diagram (Y-Z plan view) showing an example of a configuration of a radiography apparatus **10B** that is an example of a medical apparatus according to a third embodiment of the disclosed technology. In the following description, the vertical direction will be referred to as a Z direction, a travel direction of a travel portion **61** will be referred to as a Y direction, and a direction perpendicular to both the Z direction and the Y direction will be referred to as an X direction.

(115) The radiography apparatus **10B** has a so-called ceiling travel type form, and has a guide rail **60** attached to a ceiling, a travel portion **61** traveling on the guide rail **60**, a strut portion **62** that is extendable and contractible and is connected to the travel portion **61**, and a radiation source unit **13** attached to a tip end of the strut portion **62**. The radiography apparatus **10B** further includes a control unit **15** and an examination table **16**.

(116) The examination table **16** has a base portion **16A** fixed to a floor and a bed portion **16B** on which a subject (examinee) lies. A surface of the bed portion **16B** is a contact surface **17** with which the subject (examinee) comes into contact. The examination table **16** is installed to be located below the radiation source unit **13**. In a case where the travel portion **61** is traveling on the guide rail **60**, the radiation source unit **13** can be positioned, and any part of a subject (examinee) can be included in an imaging visual field. As a detection unit (not shown) for detecting radiation, a portable detection unit may be used. The detection unit may be built into the bed portion **16B**.

(117) The examination table **16** is an example of a “table portion” in the disclosed technology. The “table portion” has a contact surface with which a subject comes into contact. Each constituent of the structure including the guide rail **60**, the travel portion **61**, the strut portion **62**, and the radiation source unit **13** is an example of a “structural portion” in the disclosed technology, and is also an example of an “examination unit”. The “structural portion” is defined as a relative position with respect to the “table portion” within a predetermined range. The “examination unit” examines a subject in a non-contact manner.

(118) A sterilization unit **20** that sterilizes the contact surface **17** of the bed portion **16B** is attached to the radiation source unit **13** via an arm portion **63**. The sterilization unit **20** has an elongated rod-like shape, and one end of the sterilization unit **20** in the longitudinal direction is connected to one end of the arm portion **63**. A rotation shaft **64** having the X direction as an axial direction is provided at the other end of the arm portion **63**, and the arm portion **63** is rotatable about an axis of the rotation shaft **64**. The sterilization unit **20** is movable in the Y direction in a case where the travel portion **61** is travelling on the guide rail **60**. Similar to the first embodiment, as the sterilization unit **20**, a sterilization unit that emit light having a bactericidal action, a sterilization unit that ejects a gas or a liquid having a bactericidal action, a sterilization unit including a wiping member, and a sterilization unit having a configuration in which a plurality of types of different sterilization members are arranged may be employed.

(119) FIGS. **20A** and **20B** each are diagrams showing a state of the sterilization unit **20** in a case where the contact surface **17** is sterilized. FIG. **20A** is a Y-Z plan view, and FIG. **20B** is an X-Z plan view showing the enlarged sterilization unit **20**. In a case where the contact surface **17** is sterilized, the sterilization unit **20** is brought into the first state of being in contact with or close to

the contact surface **17** (a state in which the sterilization process can be performed). That is, as shown in FIGS. **20A** and **20B**, the arm portion **63** is positioned such that the sterilization unit **20** is located between the radiation source unit **13** and the bed portion **16B**. The entire contact surface **17** can be sterilized by causing the travel portion **61** to travel to move the sterilization unit **20** in the Y direction while setting the sterilization unit **20** to the first state. A distance between the sterilization unit **20** and the contact surface **17** can be adjusted by adjusting a length of the strut portion **62** that is extendable and contractible. In a case where the bed portion **16B** is liftable and lowerable in the Z direction, a distance between the sterilization unit **20** and the contact surface **17** may be adjusted by adjusting a height position of the bed portion **16B**.

(120) On the other hand, in a case where a radiation image is captured by the radiography apparatus **10B**, that is, in the normal use of the radiography apparatus **10B**, the sterilization unit **20** is brought into the second state (retracted state) of being separated from the contact surface **17**. That is, as shown in FIG. **19**, the arm portion **63** is positioned such that the sterilization unit **20** is located above the radiation source unit **13**.

(121) According to the radiography apparatus **10B** of the third embodiment of the disclosed technology, in the same manner as in the first embodiment, the contact surface **17** can be effectively sterilized without hindering the normal use of the radiography apparatus **10B**. In the radiography apparatus **10B**, the sterilization units **20** in the forms shown in FIGS. **9A** to **9C**, FIGS. **10A** to **10D**, and FIG. **11** may be employed.

Fourth Embodiment

(122) FIG. **21A** is a diagram (Y-Z plan view) showing an example of a configuration of a mammography apparatus **10C** that is an example of a medical apparatus according to a fourth embodiment of the disclosed technology. FIG. **21B** is a view (X-Z plan view) of the mammography apparatus **10C** as viewed from a direction of an arrow B in FIG. **21A**. In the following description, a vertical direction will be referred to as a Z direction, and a direction in which an examinee of which a radiation image is captured by the mammography apparatus **10C** faces at the time of imaging will be referred to as a Y direction, and a direction perpendicular to both the Z direction and the Y direction will be referred to as an X direction.

(123) The mammography apparatus **10C** includes an arm portion **70**, a radiation source unit **13**, a detection unit **14**, a compression plate **71**, an imaging table **72**, and a control unit **15**. The radiation source unit **13** and the imaging table **72** are attached to the arm portion **70**, and the detection unit **14** and the control unit **15** are provided inside the imaging table **72**.

(124) The arm portion **70** that is connected to a rotation shaft **73** having the Y direction as an axial direction is rotatable about an axis of the rotation shaft **73**, and is further liftable and lowerable in the Z direction. The radiation source unit **13** and the detection unit **14** can be each rotated and lifted and lowered in accordance with rotation and movement of the arm portion **70** while maintaining a positional relationship of facing each other. The compression plate **71** is provided to be liftable and lowerable in the Z direction between the radiation source unit **13** and the imaging table **72**. In a case of capturing a radiation image of a breast that is a subject, the breast is sandwiched between the compression plate **71** and the imaging table **72** and is compressed. A surface of the imaging table **72** is a contact surface **17A** with which the breast that is a subject comes into contact.

(125) The imaging table **72** is an example of a “table portion” in the disclosed technology. The “table portion” has a contact surface with which a subject comes into contact. The radiation source unit **13** is an example of a “structural portion” in the disclosed technology, and is also an example of an “examination unit”. The “structural portion” is defined as a relative position with respect to the “table portion” within a predetermined range. The “examination unit” examines a subject in a non-contact manner. The compression plate **71** is an example of a “fixing unit” in the disclosed technology. The “fixing unit” fixes a subject by sandwiching the subject between the “fixing unit” and the “table portion”.

(126) A sterilization unit **20** that sterilizes the contact surface **17A** of the imaging table **72** is

attached to the imaging table **72** via an arm portion **74**. The sterilization unit **20** is attached to the imaging table **72** via a pair of arm portions **74** attached to both ends thereof in the longitudinal direction. A roller **75** is provided at one end of the arm portion **74** in the longitudinal direction. The roller **75** is fitted in a guide groove **76** provided on the side surface of the imaging table **72** in the Y direction. The roller **75** is driven by a motor (not shown), and thus the arm portion **74** can be moved together with the sterilization unit **20** in the Y direction. The arm portion **74** is rotatable about an axis of a rotation shaft **77** having the X direction as an axial direction and provided at the same position as the attachment position of the roller **75**. Similar to the first embodiment, as the sterilization unit **20**, a sterilization unit that emit light having a bactericidal action, a sterilization unit that ejects a gas or a liquid having a bactericidal action, a sterilization unit including a wiping member, and a sterilization unit having a configuration in which a plurality of types of different sterilization members are arranged may be employed.

(127) FIGS. **22A** and **22B** are diagrams showing a state of the sterilization unit **20** in a case where the contact surface **17A** is sterilized, and are plan views corresponding to FIGS. **21A** and **21B**, respectively. In a case where the contact surface **17A** is sterilized, the sterilization unit **20** is brought into the first state of being in contact with or close to the contact surface **17A** (a state in which the sterilization process can be performed). That is, as shown in FIGS. **22A** and **22B**, the arm portion **74** is positioned such that the sterilization unit **20** is located above the imaging table **72**. By driving the roller **75** while setting a state of the sterilization unit **20** to the first state, the arm portion **74** is moved together with the sterilization unit **20** in the extension direction (Y direction) of the guide groove **76**. Consequently, it is possible to sterilize the entire contact surface **17A**.

(128) On the other hand, in a case where a radiation image is captured by the mammography apparatus **10C**, that is, in the normal use of the mammography apparatus **10C**, the sterilization unit **20** is brought into the second state (retracted state) of being separated from the contact surface **17A**. That is, as shown in FIG. **21A**, the arm portion **74** is positioned such that the sterilization unit **20** is located below the imaging table **72** at a terminal position on the side opposite to a chest wall of an examinee.

(129) According to the mammography apparatus **10C** of the fourth embodiment of the disclosed technology, in the same manner as in the first embodiment, it is possible to effectively sterilize the contact surface **17A** without hindering the normal use of the mammography apparatus **10C**.

(130) FIG. **23** is a diagram showing an example of a configuration of the sterilization unit **20** capable of simultaneously sterilizing both the contact surface **17A** of the imaging table **72** with a breast and the contact surface **17B** of the compression plate **71** with the breast. As shown in FIG. **23**, the sterilization unit **20** may include a plurality of light sources **22** having a bactericidal action and emitting light, which are arranged on both surfaces of a wiring board **21**. An optical axis direction of the light sources **22** arranged on each surface of the wiring board **21** is the Z direction. That is, the sterilization unit **20** can emit light having a bactericidal action toward both the upper side and the lower side in the Z direction. In a case where the contact surfaces **17A** and **17B** are sterilized, a state occurs in which one surface of the sterilization unit **20** faces the contact surface **17A** of the imaging table **72** with a breast, and the other surface of the sterilization unit **20** faces the contact surface **17B** of the compression plate **71** with the breast, and light from the light sources **22** is applied to both the contact surfaces **17A** and **17B**. In a case where the compression plate **71** is sterilized, it is preferable to control a height position of the compression plate **71** such that the compression plate **71** comes close to or into contact with the sterilization unit **20**.

(131) FIG. **24A** is a diagram showing another example of a configuration of the sterilization unit **20** capable of simultaneously sterilizing both the contact surface **17A** of the imaging table **72** with a breast and the contact surface **17B** of the compression plate **71** with the breast. FIG. **24B** is a sectional view taken along the line **24B-24B** in FIG. **24A**. As shown in FIG. **24B**, the sterilization unit **20** may include a plurality of light sources **22** that are arranged in the X direction and emit light having a bactericidal action, and a reflecting surface **80** that reflects light from the light source

22 to the upper side and the lower side in the Z direction. An optical axis direction of the light source **22** is the Y direction. The sterilization unit **20** may be configured to eject a liquid or a gas having a bactericidal action toward both the upper side and the lower side in the Z direction. The sterilization unit **20** may include wiping members that wipe the contact surfaces **17A** and **17B** on both one surface facing the contact surface **17A** of the imaging table **72** with the breast and the other surface facing the contact surface **17B** of the compression plate **71** with the breast.

(132) The disclosed technology can be applied to medical apparatuses other than those exemplified in the first to fourth embodiments described above. For example, the disclosed technology can be applied to a radiography apparatus of a type that captures a radiation image in an upright position, a radiography apparatus that performs fluoroscopic imaging, a magnetic resonance imaging (MRI) apparatus, a positron emission tomography (PET), a bone mineral quantification apparatus, and a radiotherapy apparatus. That is, the disclosed technology can be applied to any medical apparatus including a table portion having a contact surface with which a subject comes into contact and a structural portion in which a relative position with the table portion is determined within a predetermined range. For example, in a case where a subject comes into contact with a detector that detects radiation, such as a radiography apparatus of a type that captures a radiation image in an upright position, the detector is considered as a “table portion having a contact surface” and is a target of a sterilization process.

(133) The disclosure of Japanese Patent Application No. 2020-180888 filed on Oct. 28, 2020 is incorporated herein by reference in its entirety. All the documents, the patent applications, and the technical standards disclosed in the present specification are incorporated by reference in the present specification to the same extent as in a case where the individual documents, patent applications, and technical standards are specifically and individually stated to be incorporated by reference.

Claims

1. A medical apparatus comprising: a table portion that has a contact surface with which a subject comes into contact; a structural portion in which a relative position with the table portion is determined within a predetermined range; and a sterilization unit that is attached to the table portion or the structural portion and sterilizes the contact surface, wherein the sterilization unit is configured to switch between a first state in which the sterilization unit is in contact with or close to the contact surface and a second state in which the sterilization unit is separated from the contact surface, and wherein the structural portion is an examination unit that examines the subject in a non-contact manner.
2. The medical apparatus according to claim 1, wherein, in the first state, a distance between the sterilization unit and the contact surface is 10 cm or less.
3. The medical apparatus according to claim 1, wherein the sterilization unit includes a light source that emits ultraviolet rays.
4. The medical apparatus according to claim 1, further comprising: a radiation source unit that emits radiation, wherein the sterilization unit includes a phosphor that absorbs radiation and emits ultraviolet rays, and in the first state, the radiation radiated from the radiation source unit is applied to the phosphor.
5. The medical apparatus according to claim 1, wherein the sterilization unit ejects a liquid or a gas having a bactericidal action toward the contact surface.
6. The medical apparatus according to claim 1, wherein the sterilization unit includes a wiping member that is a member impregnated with a liquid having a bactericidal action and wipes the contact surface.
7. The medical apparatus according to claim 1, further comprising: a fixing unit that fixes the subject by sandwiching the subject between the table portion and the fixing unit, wherein the

sterilization unit is configured to emit at least one of light, a liquid, or a gas having a bactericidal action from both a first surface and a second surface opposite to the first surface, and in the first state, the first surface faces the contact surface of the table portion and the second surface faces a surface of the fixing unit that comes into contact with the subject.

8. The medical apparatus according to claim 1, further comprising: a control unit that controls a sterilization process of sterilizing the contact surface with the sterilization unit, wherein at least one of the sterilization unit or the contact surface is movable in a surface direction of the contact surface, and in the sterilization process, the control unit sets a state of the sterilization unit to the first state, and moves at least one of the sterilization unit or the contact surface in the surface direction.

9. The medical apparatus according to claim 8, further comprising: a region detection unit that detects a contact region of the contact surface with which the subject comes into contact, wherein, in the sterilization process, the control unit sterilizes the contact surface by setting a sterilization intensity in the contact region detected by the region detection unit to be higher than a sterilization intensity in a region other than the contact region.

10. The medical apparatus according to claim 8, further comprising: a height detection unit that detects a height of the contact surface, wherein, in the sterilization process, the control unit controls a height of the sterilization unit in accordance with a change in the height of the contact surface detected by the height detection unit such that a distance between the sterilization unit and the contact surface is constant.

11. The medical apparatus according to claim 8, wherein the control unit starts the sterilization process on the basis of order information regarding diagnosis or treatment of the subject.

12. The medical apparatus according to claim 8, wherein, in the sterilization process, the control unit sterilizes the contact surface in a sterilization mode selected from a plurality of sterilization modes in which sterilization intensities are different from each other.

13. The medical apparatus according to claim 1, wherein the table portion is an examination table on which an examinee to be treated in the medical apparatus lies.

14. A medical apparatus, comprising: a table portion that has a contact surface with which a subject comes into contact; a structural portion in which a relative position with the table portion is determined within a predetermined range; and a sterilization unit that is attached to the table portion or the structural portion and sterilizes the contact surface, wherein the sterilization unit is configured to switch between a first state in which the sterilization unit is in contact with or close to the contact surface and a second state in which the sterilization unit is separated from the contact surface, and wherein the structural portion is a radiation source unit that emits radiation.

15. A medical apparatus, comprising: a table portion that has a contact surface with which a subject comes into contact; a structural portion in which a relative position with the table portion is determined within a predetermined range; and a sterilization unit that is attached to the table portion or the structural portion and sterilizes the contact surface, wherein the sterilization unit is configured to switch between a first state in which the sterilization unit is in contact with or close to the contact surface and a second state in which the sterilization unit is separated from the contact surface, and wherein the structural portion is a detection unit that detects radiation.
